Rapid impairment of skeletal muscle glucose transport/phosphorylation by free fatty acids in humans.

The initial effects of free fatty acids (FFAs) on glucose transport/phosphorylation were studied in seven healthy men in the presence of elevated (1.44 +/- 0.16 mmol/l), basal (0.35 +/- 0.06 mmol/l), and low (<0.01 mmol/l; control) plasma FFA concentrations (P < 0.05 between all groups) during euglycemic-hyperinsulinemic clamps. Concentrations of glucose-6-phosphate (G-6-P), inorganic phosphate (Pi), phosphocreatine, ADP, and pH in calf muscle were measured every 3.2 min for 180 min by using 31P nuclear magnetic resonance spectroscopy. Rates of whole-body glucose uptake increased similarly until 140 min but thereafter declined by approximately 20% in the presence of basal and high FFAs (42.8 +/- 3.6 and 41.6 +/- 3.3 vs. control: 52.7 +/- 3.3 micromol x kg(-1) x min(-1), P < 0.05). The rise of intramuscular G-6-P concentrations was already blunted at 45 min of high FFA exposure (184 +/- 17 vs. control: 238 +/- 17 micromol/l, P = 0.008). At 180 min, G-6-P was lower in the presence of both high and basal FFAs (197 +/- 21 and 213 +/- 18 vs. control: 286 +/- 19 micromol/l, P < 0.05). Intramuscular pH decreased by -0.013 +/- 0.001 (P < 0.005) during control but increased by +0.008 +/- 0.002 (P < 0.05) during high FFA exposure, while Pi rose by approximately 0.39 mmol/l (P < 0.005) within 70 min and then slowly decreased in all studies. In conclusion, the lack of an initial peak and the early decline of muscle G-6-P concentrations suggest that even at physiological concentrations, FFAs primarily inhibit glucose transport/phosphorylation, preceding the reduction of whole-body glucose disposal by up to 120 min in humans.